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Spatio-temporal patterns and driving forces of surface urban heat island in Taiwan

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A R T I C L E I N F O

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Surface urban heat island (SUHI)

A B S T R A C T

The urban heat island (UHI) phenomenon, a well-documented consequence of urbanization and industrialization, is one significant anthropogenic alteration to the Earth system. The surface UHI (SUHI) has been the subject of extensive study in recent decades owing to easy access to spatially continuous satellite data observations. However, there is a lack of comprehensive SUHI studies to understand possible underlying mechanisms and drivers of SUHI's spatial variation over Taiwan. Therefore, we aim to investigate the diurnal, seasonal, and spatial patterns of SUHI intensity (SUHII) and its driving factors over eleven cities in Taiwan from 2003 to 2020. We employed Stepwise multiple regression, Pearson's correlation technique, and land surface temperature (LST) from Aqua/Terra MODIS data to explore the relationship between SUHII and its factors. Our findings reveal that the SUHII was more intense in the daytime (from 2.21 to 6.78 °C) than at night (from 0.52 to 1.63 °C), and more intensive SUHIIs were observed in northern cities (day and night: 4.99 and 1.09 °C) than in southern cities (3.35 and 1.01 °C). The SUHII exhibited significant seasonal variation, with greater variation in the day than at night. The spatial pattern of daytime SUHII was highly correlated with normalized difference latent heat index (NDLI), vegetation, built-up intensity, and anthropogenic heat emissions. In contrast, the nighttime SUHII was closely related to built-up intensity, nighttime light, and vegetation. The factors considered in this work explained a greater fraction of the SUHII variation in the daytime (79.5 to 89.0%) than at night (44.9 to 77.0%), indicating that the underlying mechanism of the spatial patterns of nighttime SUHII was more complicated, especially in spring (day vs. night: 81.5% vs. 50.3%) and winter seasons (85.3% vs. 44.9%). This study provides crucial information on the spatio-temporal patterns and driving forces of SUHI that can aid in developing heat mitigation strategies.

1. Introduction

Urbanization is one of the major anthropogenic processes altering the Earth's environment. Consequently, it has led to massive environmental issues (Grimm et al., 2008). Among them, the urban heat island (UHI), in which urban regions tend to have greater atmospheric or surface temperatures compared to surrounding rural regions, is one of the most well-documented instances of urbanization-induced impacts (Voogt and Oke, 2003; Zhou et al., 2019). Impacts of UHI may include modification of environmental conditions like biodiversity (Reid, 1998), change in vegetation activity (Imhoff et al., 2010; Yan and Zhou, 2023), reduction of the

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quality of water and air (Grimm et al., 2008; Wang et al., 2018), and influence on the local climate (Arnfield, 2003; Han et al., 2023). In addition, UHI can exert adverse effects on human well-being (Patz et al., 2005; Tan et al., 2010) and increase the demand for energy (Akbari et al., 2016; Hirano and Fujita, 2012; Liao et al., 2017). Moreover, with global climate change, these impacts are expected to be more detrimental (Li and Bou-Zeid, 2013; Patz et al., 2005; Zhao et al., 2014). Thus, a deeper understanding of the pattern and causes of the surface UHI (SUHI) effects is crucial for proposing and implementing heat mitigation strategies.

Generally, there are two main types of UHIs: atmospheric or air UHI and surface UHI (SUHI). The main difference between these UHI types is the method used to obtain the temperature (Liu et al., 2022; Yao et al., 2018a; Zhou et al., 2019). We commonly estimate air UHI from in-situ meteorological stations, and the intensity of air UHI is often calculated from the air temperature contrast between urban weather stations and rural stations located in less urbanized areas (Wang et al., 2015; Yao et al., 2018a). However, the UHI effect can influence weather stations in rural areas because of the urbanization process. Therefore, selecting the rural stations unaffected by UHI is challenging. The reason is that most meteorological stations are situated in the urban region (Wang et al., 2015). Also, the sparsely distributed meteorological station networks may only partially characterize the UHIs of a city (Zhou et al., 2014b). On the other hand, remotely-sensed land surface temperature (LST) is often utilized to compute SUHI (Firozjaei et al., 2020; Imhoff et al., 2010; Li et al., 2020b; Peng et al., 2012; Yao et al., 2018a). In recent decades, scientists have significantly studied SUHI due to open access and comprehensive coverage of satellite-based observations (Dewan et al., 2021b; Imhoff et al., 2010; Kimothi et al., 2023; Liu et al., 2022; Nguyen et al., 2018; Rasul et al., 2015; Weng et al., 2019; Xu et al., 2023).

The spatio-temporal pattern of SUHI and its associated determinants can be found in previous research on a global scale (Chakraborty and Lee, 2019; Clinton and Gong, 2013; Peng et al., 2012; Zhang et al., 2010). For instance, in their global SUHI assessment, Zhang et al. (2010) only concentrated on examining the SUHI during the daytime and the ecological control of global variations. Using 419 major cities worldwide, Peng et al. (2012) reported that the daytime SUHII was greater than nighttime SUHII and the urban-suburban contrast in vegetation activity (in the day) and in surface albedo and anthropogenic emissions (during the nighttime) caused their spatial variations. Chakraborty and Lee (2019) highlighted that SUHII's diurnal variation was greatest in equatorial climates and lowest in arid regions, while Li et al. (2020a) conducted globally on >400 cities and stated that driving forces associated with SUHII differed spatially among cities.

Similar studies were also implemented at a regional scale (Dewan et al., 2021b; Li et al., 2022; Mohammad and Goswami, 2021; Xin et al., 2022; Zhang et al., 2023). For example, using 38 populous cities in the US, Imhoff et al. (2010) reported a greater daytime SUHII compared to that at night, higher SUHII in summer compared to winter, and SUHII was primarily influenced by ecological context across cities. Zhou et al. (2014b) conducted a study on 32 big cities in China to investigate the spatial patterns and drivers of SUHI. The study showed that SUHII demonstrated obvious spatial-temporal patterns, and climate was the primary factor affecting SUHII variation across cities. Additionally, the daytime SUHII had more complicated underlying mechanisms than at night. Mohammad and Goswami (2021) implemented a study to quantify the diurnal and seasonal variation of SUHII and its associated determinants over 150 big cities in India located across various climatic regions. Their findings indicated that SUHII exhibited more significant variation during daytime compared to nighttime (except for tropical cities), and the SUHII patterns for cool cities can be better understood than hot cities. In addition, this work quantified the substantial influence of thermal inertia and soil moisture in comprehending urban heat and cool islands across various climatic zones. Dewan et al. (2021b) examined the diurnal and seasonal SUHI variation, temporal trends, and possible driving forces in five large cities in Bangladesh. They found that SUHII was higher in the daytime than during the nighttime, and the factors controlling SUHII were population, lack of greenness, and anthropogenic forcing. Moreover, they reported an increased trend of daytime SUHII for the five cities of concern over time. While these studies have expanded our understanding of spatiotemporal dynamics of SUHI, the utilization of various data sources, quantification techniques, associated determinants, and varying urban boundary delineations can introduce uncertainty (Manoli et al., 2019) and lead to highly diverse conclusions (Zhou et al., 2014b), especially when investigating densely populated and rapidly developing urban areas (Dewan et al., 2021a). Consequently, the applicability of these findings at the local level becomes challenging (Dewan et al., 2021a).

Previous work used a range of different driving forces associated with SUHII, such as enhanced vegetation index (EVI), normalized difference vegetation index (NDVI) (Imhoff et al., 2010; Shastri et al., 2017), evapotranspiration (ET) (Shastri et al., 2017; Liou and Kar, 2014), white sky albedo (WSA) (Peng et al., 2012; Zhou et al., 2014b), impervious surface area (ISA) (Zhang et al., 2010), climate variables (Zhou et al., 2016b; Zhou et al., 2014b), and other indicators. Recently, we developed the Normalized Difference Latent Heat Index (NDLI) (Le and Liou, 2021a, 2021b; Liou et al., 2018), which served as an indicator to describe the potential latent heat flux of the Earth's surface. Because evaporation and transpiration from water surfaces and vegetation can generate a cooling effect on LST, our research considered NDLI as a potential contributing factor in comprehending SUHI, which has not been previously explored in the literature of SUHI studies worldwide. Thus, this study is the first effort to apply the newly developed NDLI to study the SUHI.

Preceding research implemented in Taiwan have focused mostly on the air UHI effects (Chao et al., 2021; Huang et al., 2020; Lin et al., 2008; Lin et al., 2011; Sun et al., 2019), but less attention has been paid to study SUHI phenomenon. As far as we know, in Taiwan, SUHI studies have been mostly implemented in one or several big cities and are primarily interested in the magnitude of SUHI rather than the diurnal, seasonal, and spatial patterns of SUHI and its relationships with possible driving forces (Kuo et al., 2013; Lu et al., 2017; Wu et al., 2013). For example, Wu et al. (2013) studied the impacts of SUHI under heatwave influences in two northern Taiwanese cities (Taipei metropolis and Yilan) using MODIS-derived LST and three-dimensional Urbanization Index (3DUI). Liao (2008) studied the seasonality of the SUHI in four major Taiwanese cities employing MODIS data. However, the study had a short timeframe (from 2003 to 2005) and only investigated the relationship between temperature and several image-based indices. Hence, these studies have not sufficiently addressed the in-depth analysis of spatio-temporal SUHI patterns and drivers for Taiwanese cities. Consequently, there is a pressing need for a comprehensive study to investigate the diurnal, seasonal, and spatial patterns of SUHI and its associated factors in Taiwan.

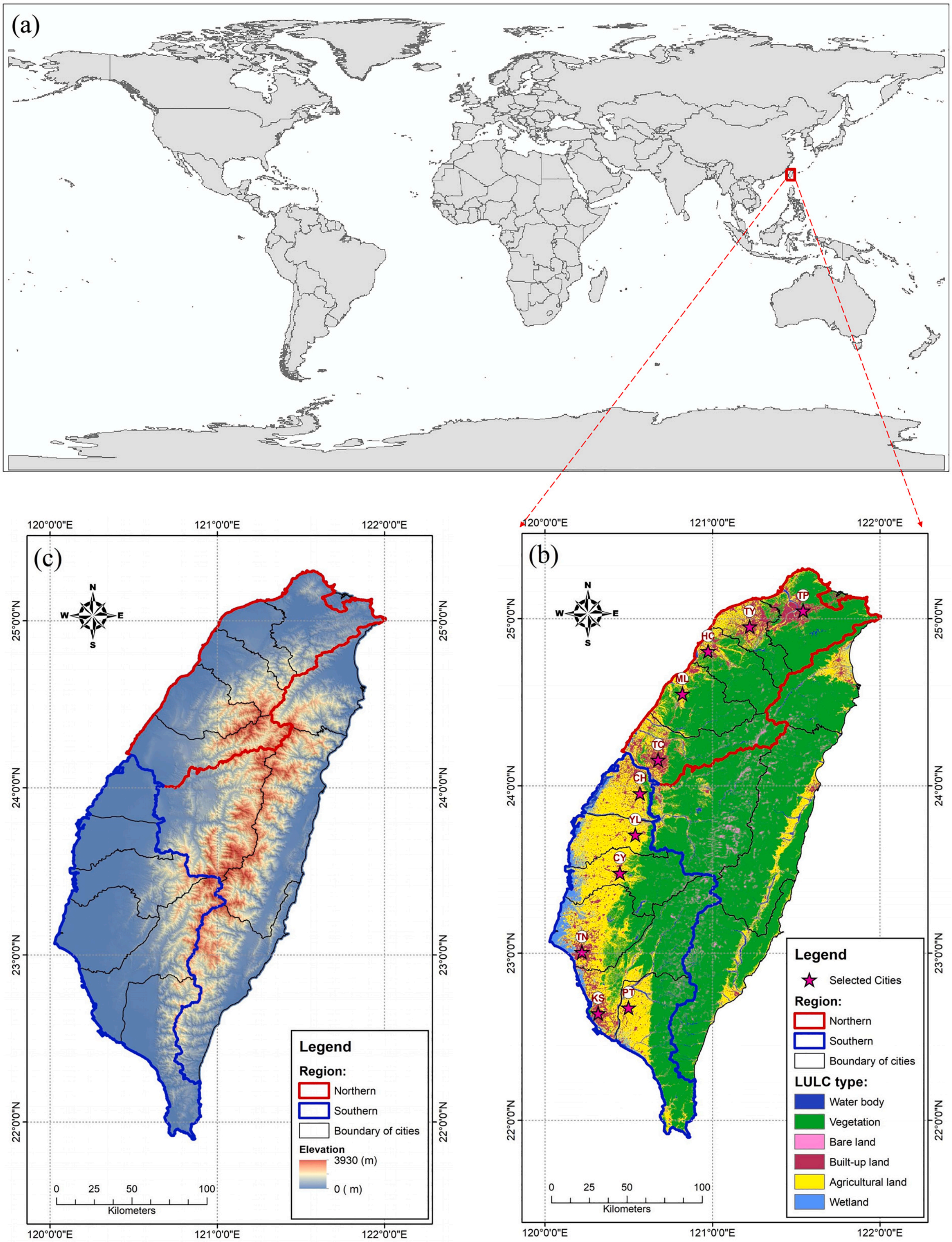

Fig. 1. (a) Geographical location of Taiwan in the world; (b) Location of the eleven selected Taiwanese cities (the LULC acting as the background map); and (c) Digital Elevation Model (DEM) of Taiwan. ArcGIS was used to create the maps.

This study was implemented to help fill the research gaps mentioned above. Additionally, we conducted a comprehensive study of spatio-temporal SUHII patterns for Taiwan and applied a newly developed NDLI as a potential contributing SUHI factor. Specifically, we employed the MODIS LST data to compute the LST difference between urban and rural regions for eleven Taiwanese cities. Also, stepwise multiple regression and Pearson correlation methods were implemented to examine the associated determinants for the spatial patterns of SUHI intensity (SUHII). This research aims (1) to characterize the diurnal, seasonal, and spatial patterns of SUHII for eleven cities and (2) to identify their possible influential factors.

2. Study area and datasets

2.1. Study area

This study examines SUHII in eleven Taiwanese cities, as shown in Fig. 1. It is important to note that Taipei comprises Taipei City and New Taipei City. At the same time, Chiayi includes both Chiayi City and Chiayi County, and Hsinchu encompasses both Hsinchu City and Hsinchu County. The eleven chosen cities are all situated in the western part of Taiwan. This is because obtaining good-quality images from central and eastern Taiwan was difficult due to cloud effects throughout the whole year. In addition, the eleven cities were divided into northern and southern regions, considering their geographical locations and land use/land cover (LULC) types to investigate the spatial variability of SUHII. Specifically, the northern region includes Taipei (TP), Taoyuan (TY), Hsinchu (HC), Miaoli (ML), and Taichung (TC), whereas the southern region comprises Changhua (CH), Yunlin (YL), Chiayi (CY), Tainan (TN), Kaohsiung (KS), and Pingtung (PT) (see Fig. 1). Table 1 shows some basic statistic information of the 11 selected Taiwanese cities for the year 2020.

2.2. Datasets and data preprocessing

The overall flowchart of the research method in this study have been demonstrated in Fig. 2. In this work, the data were obtained from several sources/products as listed in Table 2. For example, remote sensing data were preprocessed and downloaded from Google Earth Engine platform (Gorelick et al., 2017) (https://developers.google.com/earth-engine/datasets/catalog). Additionally, air temperature and precipitation data were downloaded from Central Weather Bureau (CWB) (https://e-service.cwb.gov.tw/HistoryDataQuery/index.jsp). To characterize the SUHI phenomenon, we used MODIS-derived LSTs data from 2003 to 2020. For each city, the LST information was obtained from products of MYD11A2-Aqua/MOD11A2-Terra MODIS with 8-day intervals, 1-km resolution, and high-quality control. The data were available at four different times a day, including 01:30 (nighttime-Aqua), 10:30 (daytime-Terra), 13:30 (daytime-Aqua), and 22:30 (nighttime-Terra). These MODIS-derived LST products were retrieved from clear-sky observations (with 99% confidence) utilizing a universal split-window LST algorithm. These products have also been widely validated using in-situ data from many test regions with better than one-degree accuracy (Wan, 2008, 2014; Wan and Dozier, 1996). Also, noises caused by cloud effects, topographical distinctions, and changes in zenith angle were further processed to improve the accuracy of LST retrieval (Wan, 2008, 2014). In Taiwan, due to frequent afternoon clouds of the subtropical weather type, it is challenging to obtain cloud-free images in summer (Wu et al., 2013; Wu et al., 2010). Therefore, low-quality LST values due to cloud issues or other processing failures were filtered out using the quality control (QC) values of LST products. At the same time, only good-quality LST data (LST error $\leq$ 1 K) were retained for further analysis.

The present study also used other categories of MODIS products. Previous research has indicated that the Enhanced Vegetation Index (EVI) is the more controlling driver of SUHI than the Normalized Difference Vegetation Index (NDVI) (Zhou et al., 2014b). Thus, to reflect the vegetation condition of each city, the 16-day interval MODIS EVI (version-6) (MOD13A2) with a spatial resolution of 1 km during 2003–2020 was incorporated in our work. The QC information of the EVI product was employed to filter out only those pixels with good quality, which were utilized for further analysis. In addition, a Savitzky–Golay filtering technique (Chen et al., 2004) was applied to remove further possible noises resulting from cloud effect, atmospheric variability, and bi-directional impacts.

As for albedo data, Peng et al. (2012) suggested that shortwave black sky albedo (BSA) had a linear correlation with white sky

Table 1
Statistic information of the 11 selected Taiwanese cities for the year 2020.

City	Region	Area (Km^2)	No. of Household	Population (Persons)	Population Density (Persons/km^2)
Taipei (TP)	Northern	2324	2,666,315	6,638,252	2856
Taoyuan (TY)	Northern	1221	843,925	2,265,940	1856
Hsinchu (HC)	Northern	1532	375,617	1,020,358	666
Miaoli (ML)	Northern	1820	193,141	543,287	298
Taichung (TC)	Northern	2215	999,909	2,817,547	1272
Changhua (CH)	Southern	1074	396,310	1,267,786	1180
Yunlin (YL)	Southern	1291	243,900	677,669	525
Chiayi (CY)	Southern	1964	285,743	766,368	390
Tainan (TN)	Southern	2192	703,073	1,875,909	856
Kaohsiung (KS)	Southern	2952	1,118,827	2,767,383	938
Pingtung (PT)	Southern	2776	291,522	813,676	293

Source: Dept. of Household Registration Affairs, MOI. (https://www.ris.gov.tw/app/en).

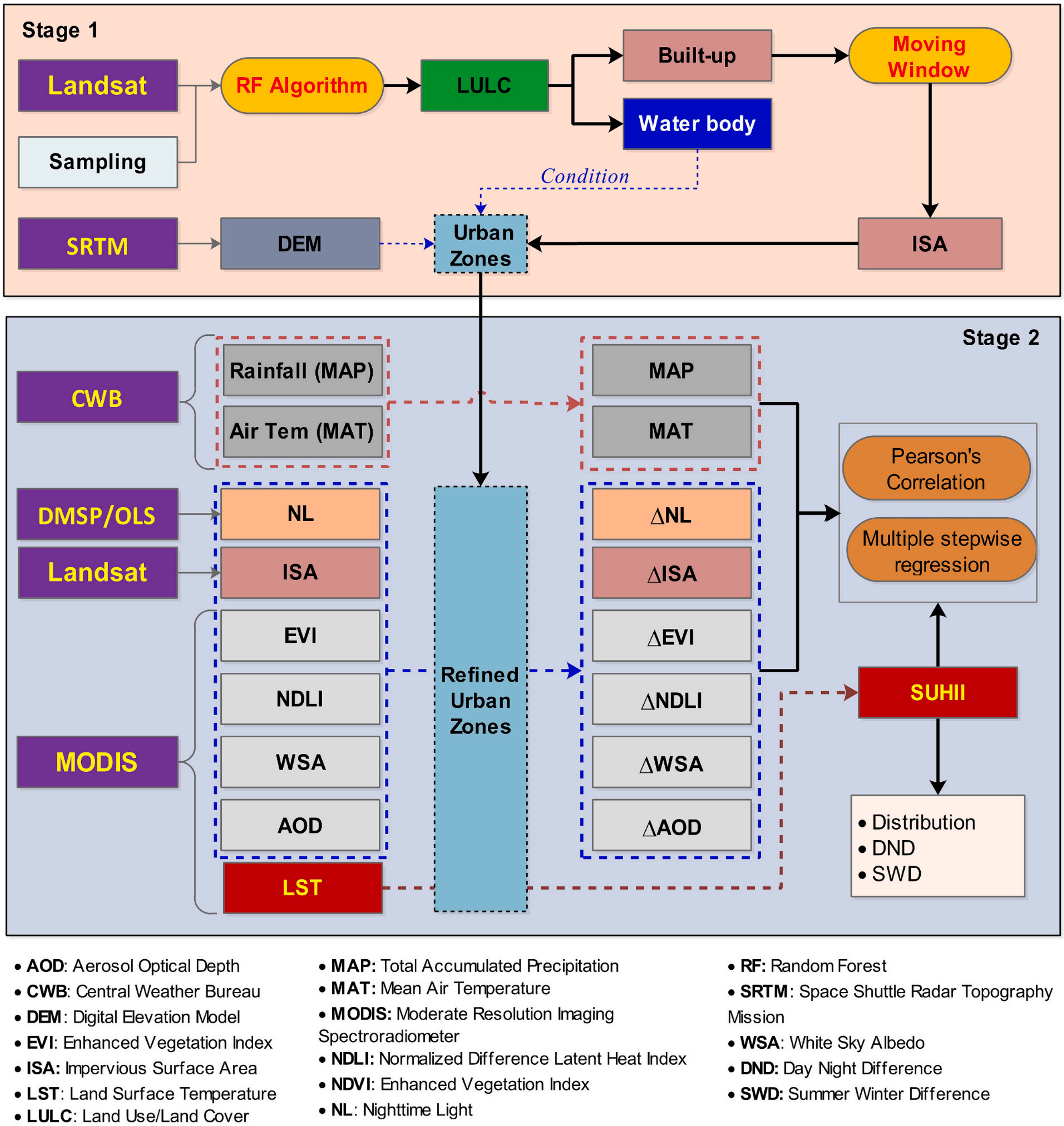

Fig. 2. Overall flowchart of research method.

albedo (WSA) and exhibited the sample relationship with SUHII to WSA. Thus, we only utilized WSA in this work. The WSA information was derived from a 16-day composite MODIS albedo product (MCD43A3) at 1-km resolution. Moreover, to characterize each city's Aerosol Optical Depth (AOD) information, the level 2 MODIS AOD product named MCD19A2 at 0.55 μm (daily composite, 1-km resolution) was used. For both WSA and AOD, a quality band was used as a filter to retain high-quality pixels.

In this study, NDLI served as an indicator to characterize the potential latent heat flux of the Earth's surface. The computation of NDLI relied on surface reflectance data (MOD09A1) and used the Eq. (1) (Le and Liou, 2021a, 2021b; Liou et al., 2018):

$$\mathrm{NDLI} = \frac{\mathrm{Green\text{-}Red}}{\mathrm{Green} + \mathrm{Red} + \mathrm{SWIR}} \tag{1}$$

where Red, Green, and SWIR are band 1, band 4, and band 6 of the MODIS spectral reflectance (MOD09A1), respectively. NDLI has the potential to serve as a contributing factor to SUHI, an aspect that has yet been explored in SUHI research.

Table 2
Data description and sources of data.

Derived variables	Data description	Products	Sources
LST	8-day, at 1 km, 2003–2020	MOD11A2 (MODIS), MYD11A2 (MODIS)	Link1 Link2
EVI/NDVI	16-day, at 1 km, 2003–2020	MOD13A2 (MODIS)	Link3
WSA	16-day, at 1 km, 2003–2020	MCD43A3 (MODIS)	Link4
AOD	daily, at 1 km, 2003–2020	MCD19A2 (MODIS)	Link5
NDLI	8-day surface reflectance (SR), at 0.5 km, 2003–2020	MOD09A1 (MODIS)	Link6
30-m LULC	SR, at 30 m, years (2005, 2010, 2015, and 2020)	TM/OLI (Landsat)	Link7
DEM	DEM data, at approximately 90 m	SRTM	Link8
Nighttime light	Annual data, at 1 km, years (2005, 2010, and 2013)	DMSP/OLS	Link9
Air temperature	Daily air temperature, 2003–2020	Stations	Link10
Precipitation	Daily precipitation, 2003–2020	Stations	Link10

Link: https://developers.google.com/earth-engine/datasets/catalog;
Link1: Link + /MODIS_061_MOD11A2;
Link2: Link + /MODIS_061_MYD11A2;
Link3: Link + /MODIS_061_MOD13A2;
Link4: Link + /MODIS_061_MCD43A3;
Link5: Link + /MODIS_061_MCD19A2_GRANULES;
Link6: Link + /MODIS_061_MOD09A1;
Link7: Link + /landsat;
Link8: Link + /CGIAR_SRTM90_V4;
Link9: Link + /NOAA_DMSP-OLS_NIGHTTIME_LIGHTS;
Link10: https://e-service.cwb.gov.tw/HistoryDataQuery/index.jsp

This study utilized the Version 4 data of remotely sensed stable nighttime lights (NL) from the Defense Meteorological Satellite Program - Operational Line scan System (DMSP/OLS) as a surrogate of anthropogenic heat emissions following previous works (Mohammad and Goswami, 2021; Peng et al., 2012; Zhang et al., 2010) to gain knowledge about the influence of anthropogenic forcing. The DMSP/OLS data with a 30-arc-second (about 1 km) resolution were collected annually. Additionally, a Digital Elevation Model (DEM) from the NASA Space Shuttle Radar Topography Mission (SRTM) was employed as a filter to eliminate the elevation impact on computing SUHII, in line with previous research (Imhoff et al., 2010; Yao et al., 2018a) (Fig. 1 and Fig. 5).

For climate data, daily rainfall and air temperature were downloaded from the Central Weather Bureau (CWB) from 2003 to 2020 to reflect the characteristics of cities' climatic conditions. In this study, these in-situ weather station networks are inside their suburban or urban areas, which may cause some biases because of possible urbanized and topographical impacts. However, they may not significantly change cities' spatial background climate patterns (Arnfield, 2003; Zhou et al., 2016b).

LULC maps were created for each city within their administrative borders using Landsat-derived (TM/OLI) satellite data with a 30 m spatial resolution for four years (2005, 2010, 2015, and 2020). The process involved four main steps. Firstly, the Landsat-derived data were preprocessed, including re-projection and mosaic. Secondly, training samples were carefully selected using the knowledge of experts with the aid of high-resolution images from Google Earth (GE). The training samples were divided into the training part (70%) and the testing part (30%). Thirdly, the training part was employed to classify Landsat satellite images into five classes (waterbody, forest, agricultural land, built-up land, bare land, and wetland) employing a random forest (RF) algorithm (Breiman, 2001; Liou et al., 2022; Talukdar et al., 2020). Finally, the testing part was used to evaluate the accuracies of the LULC maps. Using the Gird Search

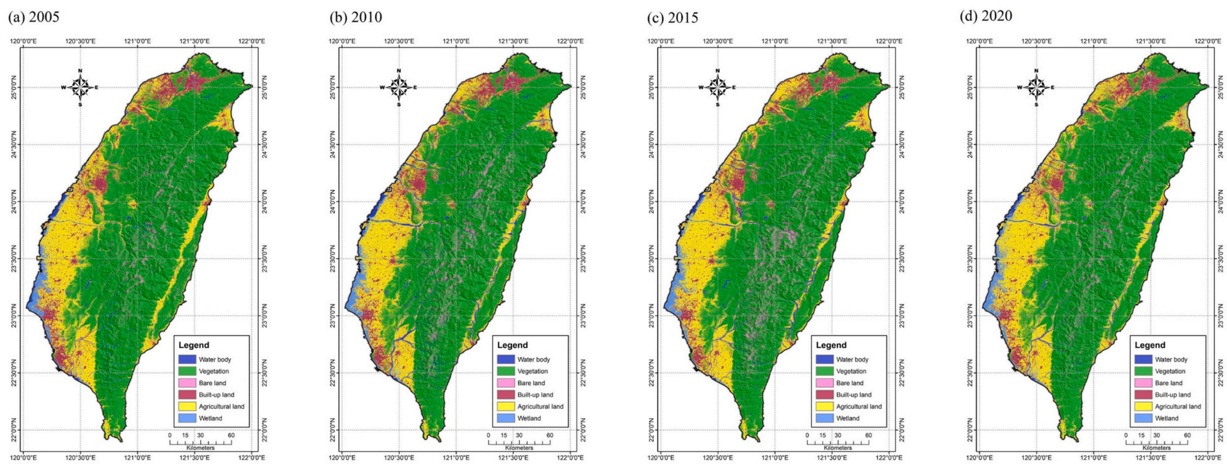

Fig. 3. LULC maps using RF model for different years: (a) 2005; (b) 2010; (c) 2015; and (d) 2020.

technique, we tuned two important hyperparameters of the RF algorithm: *n_estimators* (the number of trees in the forest) and *max_depth* (the tree's maximum depth). Specifically, we tuned the tree's maximum depth (*max_depth)* from a list of values (1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, and 15). Also, the *n_estimators* value was searched over a list of values: 1, 3, 5, 8, 12, 16, 20, 30, 40, and 60. After considering the model performance and computational time, we set the following hyperparameters: *n_estimators* = 30, *max_depth* = 12 for the RF model used for LULC classification. This work employed the Python scikit-learn package to build and execute the RF algorithm (Pedregosa et al., 2011). A desktop computer (with an Intel (R) Core (TM) i7–9700 CPU, 3.00GHz with 16 cores and 64GB RAM) was utilized to run the RF algorithm. Accuracy assessment results show that the accuracy (Kappa index) values are 0.952 (0.948), 0.966 (0.963), 0.957 (0.953), and 0.974 (0.972) for 2005, 2015, 2015, and 2020 respectively, which fulfilled the accuracy criteria for this analysis. Fig. 3 shows the LULC maps of Taiwan for four different years: 2005, 2015, 2015, and 2020.

Based on the LULC map, we extracted built-up intensity (BI) or Impervious Surface Area (ISA) information to delineate the rural and urban areas (Section 3.2). After having all the factors influencing SUHI, we demonstrated the spatial patterns of determinants associated with SUHI at an annual scale (Fig. 4).

3. Methodology

3.1. Different ways of SUHII calculation

Numerous methodologies exist in the literature for computing Surface Urban Heat Island Intensity (SUHII). The choice of SUHI definition significantly impacts the accurate quantification of the magnitude and spatial pattern of the SUHI effect (Schwarz et al., 2011; Zhao et al., 2016; Zhou et al., 2015). For example, Schwarz et al. (2011) conducted a comprehensive analysis comparing eleven different SUHIIs (indicators) commonly utilized in remotely sensed-based UHI assessments across 263 European cities. Using monthly mean LST from MODIS for July 2002, January 2003, and July 2003, their study yielded significant findings. Their results highlighted two crucial observations: Firstly, despite aiming to characterize the same SUHI phenomenon, the different indicators displayed weak correlations when assessing single points in time. Secondly, these indicators revealed daily and seasonal patterns individually, but they

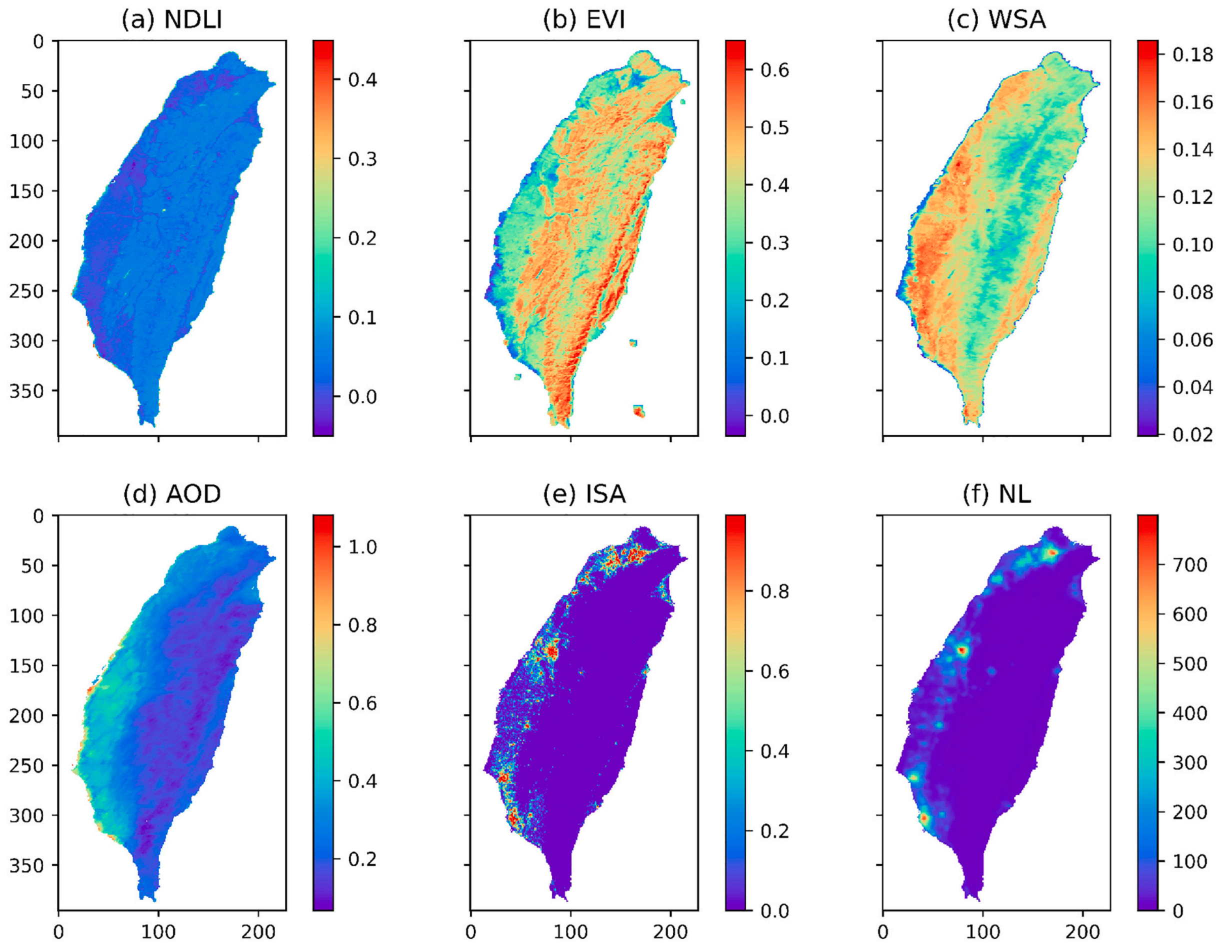

Fig. 4. Spatial patterns of determinants associated with SUHI at an annual scale for Taiwan.

demonstrated relatively low correlations over time. Consequently, the study emphasized that researchers must meticulously consider the disparities and inconsistencies among selected indicators when investigating SUHI.

Selecting reliable rural references when calculating SUHII is crucial. Nevertheless, the approaches for delineating rural areas differed significantly across various studies. Numerous research computed the SUHII by measuring the LST contrast between urban and adjacent suburban regions (Clinton and Gong, 2013; Mohammad and Goswami, 2021; Peng et al., 2012; Shastri et al., 2017; Zhao et al., 2016; Zhou et al., 2014b). At the same time, many studies computed the SUHII as an urban-rural LST difference (Imhoff et al., 2010; Yao et al., 2017; Zhang et al., 2010; Zhou et al., 2016b). In addition, elevation and water body significantly influence SUHII calculation, and we may need to exclude them when computing SUHII, but many previous studies have ignored the steps. To investigate these issues, Yao et al. (2018a) implemented a study to explore the effects of employing distinct methodologies to define rural areas and examine the impact of altitude and water bodies on computing SUHII across 31 Chinese cities. Their findings revealed that, on summer days (SDs), neglecting the impact of altitude and water bodies would result in an overestimation of the SUHII by an average of 1.68 °C (across 31 cities) and 0.28 °C, respectively. Additionally, employing the nearby suburban area as a reference led to an underestimation of SUHII by 1.48 °C during SDs.

Therefore, in our scenario, after carefully considering different SUHII definitions and based on the suggestion provided by Yao et al. (2018a) in consideration of the characteristics of Taiwanese cities, we adopted the definition of SUHII as the contrast in LST between urban and rural. This definition stands as one of the most commonly used and traditional approaches in SUHI assessment, extensively employed in prior studies for computing SUHII within individual cities or cross-city comparisons (Oke, 1982; Stewart, 2011; Zhou et al., 2016a). Nevertheless, the distinction between what constitutes "urban" and "rural" remains multifaceted and can lead to confusion (Schwarz et al., 2011; Stewart and Oke, 2009). While rural areas are defined as parts of a city not impacted by the UHI (Lowry, 1977), remotely sensed studies often rely on predetermined definitions of "urban" and "rural" areas based on the land cover data. Consequently, scientists have employed various methodologies to delineate these two areas (Schwarz et al., 2011). Given the relatively small area of Taiwanese cities compared to larger global cities (Table 1), this study followed the methodology established by Zhou et al. (2016a) to delineate urban and rural areas (Section 3.2).

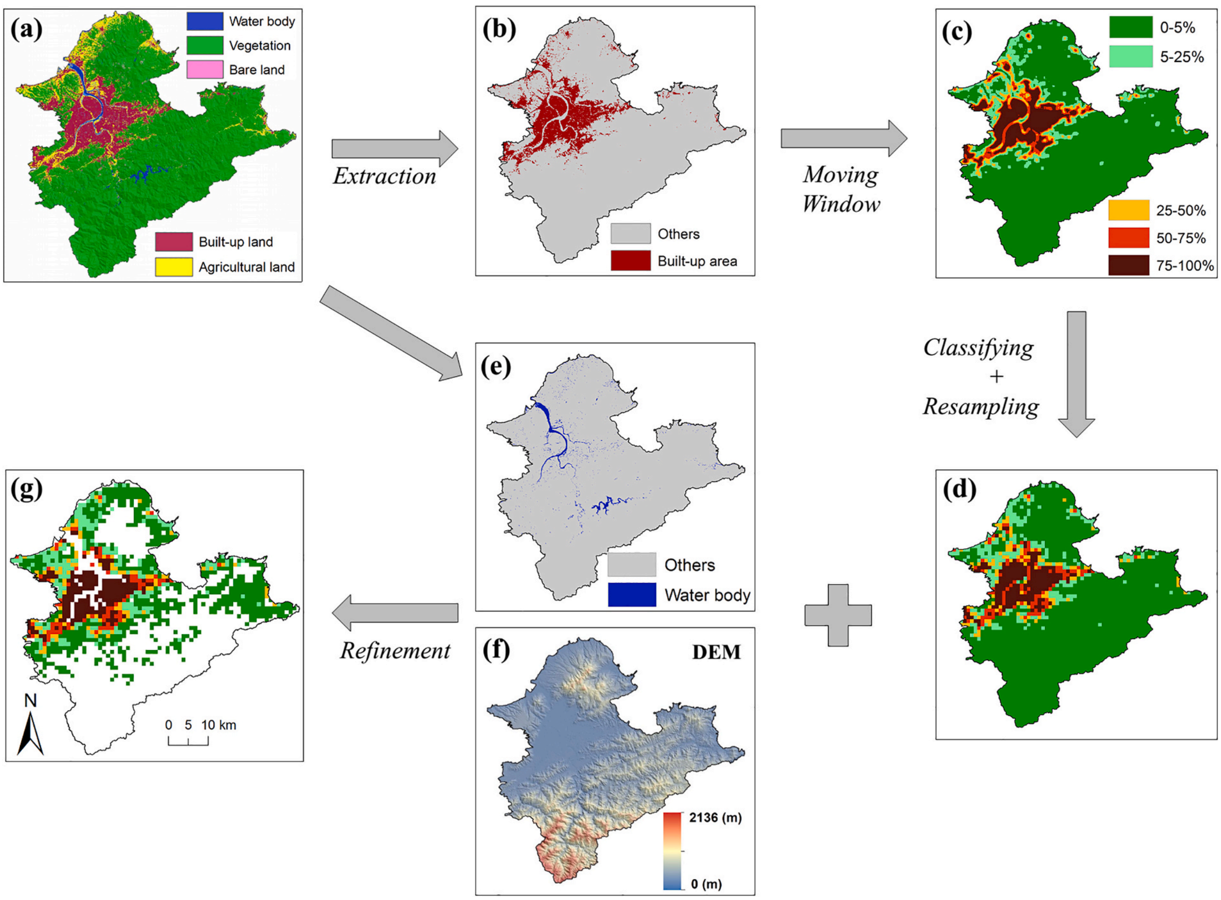

Fig. 5. The flowchart of delineation of urban and reference rural regions, Taipei as an example. (a) LULC map in 2020 (30 m), (b) built-up land map, (c) built-up intensity (BI) map, (d) classified zones with different built-up intensity (BI) map, (e) water body map, (f) Digital Elevation Model (DEM), and (g) zones after refinement (1 km). On the map, the blank regions were not used in further analysis.

3.2. Delineation of urban and rural areas

In this study, five steps were conducted to define urban and rural areas to provide a base for SUHII computation (Fig. 5):

1) We extracted built-up areas and water bodies from the LULC maps derived from Landsat images with a resolution of 30 m (Fig. 5(b), 5(e)).
2) We used a moving window algorithm of 33 × 33 pixels based on the built-up area map to produce the built-up intensity (BI) map (Zhou et al., 2014b) (Fig. 5(c)).
3) Based on BI thresholds, we initially classified the BI map into five zones, including rural (BI ≤5%) and four urban zones (exurban (5% < BI ≤25%), suburban (25% < BI ≤50%), urban (50% < BI ≤75%), and urban core (75% < BI ≤100%)) (Fig. 5(d)) (Zhou et al., 2016a). The classified BI map (30 m) was then resampled (using the nearest neighbor algorithm) to a 1-km resolution to match the MODIS-derived LST pixel size.
4) A 50% threshold of the BI criterion was applied to classify the BI map into low BI and high BI groups (Imhoff et al., 2010). After that, the final urban area in this study was delineated by aggregating the high BI polygons.
5) We removed water body pixels and rural pixels that were situated at elevations 50 m higher than the maximum altitudes of the urban regions because they may make the effects of urbanization on LST less important (Imhoff et al., 2010; Li et al., 2020b; Zhou et al., 2014a) (Fig. 5(e) and Fig. 5(f)).

In this work, the boundary of urban and rural areas was extracted from the Landsat-derived LULC maps for four specific years (2005, 2010, 2015, and 2020) for each city separately. The reason is that according to Zhao et al. (2016), utilizing dynamic urban coverage and LST maps is crucial to quantify UHI because employing outdated urban coverage maps to define an urban and rural area may cause biases in SUHI analysis. Prior studies used LULC maps of some specific years to represent the boundary of zones in a relatively short time frame to reduce the biases caused by out-of-date urban zone maps (Zhou et al., 2016a; Zhou et al., 2014b). For instance, Zhou et al. (2014b) established urban and suburban regions on city maps spanning 2000, 2005, and 2010 (covering from 2003 to 2011). They posited that the delineations of urban and suburban areas in those specific years could depict the corresponding areas in the intervening periods: 2003–2004, 2005–2009, and 2010–2011, respectively. Similarly, Zhou et al. (2016a) examined the SUHII in conjunction with Urban Development Intensity (UDI) between 2003 and 2012, and they made the assumption that the UDI maps from 2005 and 2010 could be used to represent the years 2003–2007 and 2008–2012, respectively.

In line with other studies, during the SUHII investigation, we did not maintain a consistent urban and rural boundary from 2003 to 2020. Instead, we delineated urban and rural areas using LULC maps of four years (2005, 2010, 2015, and 2020), which can reduce the effects of LULC changes or errors caused by out-of-date urban zone maps on SUHII calculation. More specifically, we assumed that the urban and rural areas identified in 2005, 2010, 2015, and 2020 could represent those for 2003–2007, 2008–2012, 2013–2017, and 2018–2020, respectively (Fig. A1). Additionally, we selected the 5-year interval between LULC maps because Taiwan's LULC change is not very significant, and other scientists also accepted the 5-year interval of LULC maps (Zhou et al., 2016a; Zhou et al., 2014b).

3.3. Computing of SUHII and its potential driving forces

As mentioned before, the intensity of SUHI (SUHII) was quantified as the urban-rural difference in LST (Imhoff et al., 2010; Yao et al., 2018a) and the delineation of urban and surrounding rural regions in Section 3.2. The SUHII computing was carried out using Eq. (2):

$$SUHII = LST_{urban} - LST_{rural} \tag{2}$$

where the LST_{rural} and LST_{urban} are the average satellite-derived LST for the rural and urban pixels, respectively.

For each city, the SUHII in the daytime (10:30 and 13:30) and nighttime (22:30 and 01:30) were calculated annually and seasonally (spring: March, April, and May; summer: June, July, and August; autumn: September, October, and November; and winter: December, January, and February) for the period spanning from 2003 to 2020. Also, for each city, the day-night difference (DND) (computed as SUHII contrast between daytime and nighttime) and summer-winter difference (SWD) (calculated as subtracting SUHII in summer by SUHII in winter) were estimated to characterize the magnitudes of diurnal and seasonal SUHII (Zhou et al., 2016b; Zhou et al., 2014a).

Along with the calculation of SUHII, six urban-rural contrast of factors or drivers (referred to as delta 'Δ'), including Enhanced Vegetation Index (ΔEVI), Aerosol Optical Depth (ΔAOD), White Sky Albedo (ΔWSA), Normalized Difference Latent Heat Index (ΔNDLI), Built-up Intensity (ΔBI or ΔISA), and Nighttime Lights (ΔNL), were also computed using the same way as in Eq. (2). These six variables and two climate variables (average air temperature (MAT) and total accumulated precipitation (MAP)) were analyzed to identify the possible driving forces contributing to the spatial variabilities of SUHII. While we computed the annual and seasonal information of ΔWSA, ΔAOD, ΔEVI, ΔNDLI, MAT, and MAP for each city during the timeframe 2003–2020, ΔNL and ΔBI were calculated only for the annual scale due to no seasonal information available for them in this present study's dataset.

3.4. Investigation of the relationship between SUHII and its driving forces

In this work, stepwise multiple regression analysis, as applied in other research (Mohammad and Goswami, 2021; Peng et al., 2012), was performed to explore the potential associated driving forces with SUHII. The goal was to identify the most significant explanatory variables and determine their independent explanation rate to the geographic variation of SUHII. The dependent variables in our regression model were diurnal and seasonal SUHIIs, while the independent variables included eight factors: ΔEVI, ΔWSA, ΔAOD, ΔNDLI, ΔBI, ΔNL, MAT, and MAP. We also used Pearson's correlation analysis to estimate the correlation coefficient values between the aforementioned factors and SUHII among eleven cities. This work performed these analyses on Statistical Product and Service Solutions (SPSS) software.

4. Results

4.1. Seasonal and diurnal characteristics of SUHII

Fig. 6 demonstrates the spatiotemporal variations of SUHII averaged during 2003–2020, both seasonally and diurnally, in eleven cities in Taiwan. Noticeably, only positive SUHIIs were witnessed day or night in all seasons and all eleven Taiwanese cities, without any urban cool island (UCI) phenomena.

4.1.1. Diurnal variations of SUHII

The mean annual daytime/nighttime SUHI intensities averaged for 18 years (2003−2020) for each city are shown in Figs. 6(a) and 6(g). The results show positive values of SUHII for all eleven cities in Taiwan, with daytime SUHII ranging from 2.21 (Changhua) to

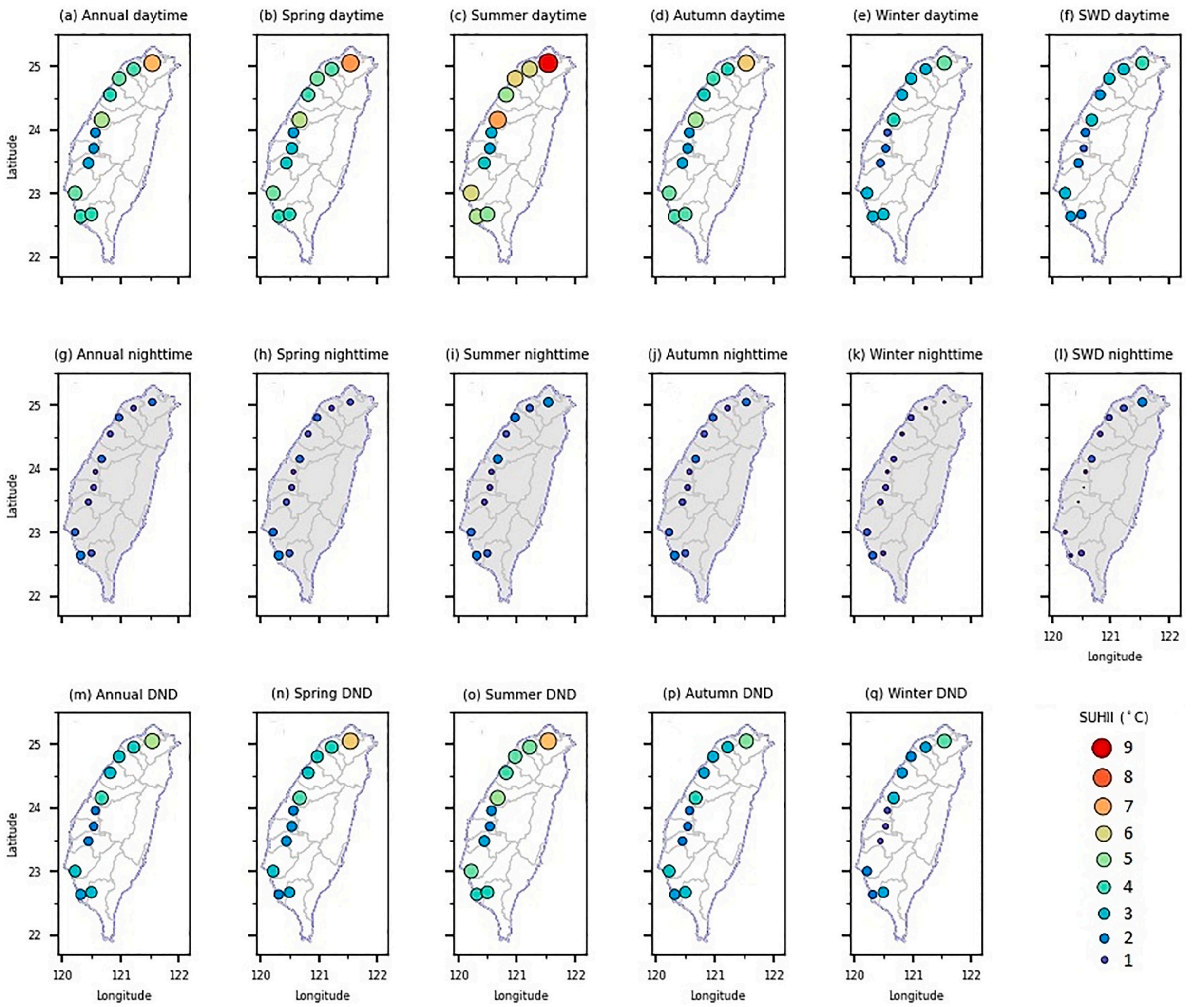

Fig. 6. Spatial distributions of SUHII in eleven cities of Taiwan averaged over 2003–2020, including daytime SUHII for (a) annual, (b) spring, (c) summer, (d) autumn, and (e) winter; nighttime SUHII for (g) annual, (h) spring, (i) summer, (j) autumn, and (k) winter; day-night difference of SUHII (DND) for (m) annual, (n) spring, (o) summer, (p) autumn, and (q) winter; and the summer-winter difference (SWD) (f) in the day and (l) at night.

6.78 °C (Taipei) and nighttime SUHII varying from 0.52 (Changhua) to 1.63 °C (Kaohsiung). When averaging over all the eleven Taiwanese cities, the annual SUHII varied substantially in a diurnal cycle, with greater SUHII in the daytime (4.10 ± 1.33 °C, mean ± standard deviation) compared to the nighttime (1.05 ± 0.35 °C).

The diurnal cycles of annual SUHII for both the Taiwan regions are demonstrated in Fig. 7. For the northern region of Taiwan, the annual SUHII in the daytime was more significant as compared with nighttime SUHII, with average values of SUHII varying from 1.09 ± 0.31 °C at 01:30 to 4.99 ± 1.14 °C at 13:30, respectively (Fig. 7). The diurnal pattern of the southern region of Taiwan was similar but with a smaller variation (from 1.01 ± 0.40 °C at 01:30 to 3.35 ± 1.02 °C at 13:30, respectively) (Fig. 7).

4.1.2. Seasonal variations of SUHII

The seasonal variations of SUHIIs over eleven Taiwanese cities are shown in Fig. 6 and Fig. 8. In the daytime, the SUHIIs of eleven cities witnessed significant seasonal variations (Fig. 6). Overall, during the daytime, most of the cities (90.9%, Fig. 8(b)) exhibited the greatest SUHIIs in summer, except Yunlin County, where the SUHII maximized (3.14 °C) in the spring season (Fig. 8(a1)). In addition, all eleven cities of Taiwan had the smallest daytime SUHIIs in winter (Fig. 8(a2)). However, at night, while the weakest SUHIIs were found in most Taiwanese cities in winter (except Yunlin in summer) (Fig. 8(a4)), the strongest SUHIIs occurred in summer in all the northern cities of Taiwan and Pingtung (54.5%), but in spring for most of the southern counterparts (36.4%).

Fig. 9 shows the SUHIIs in the daytime (at 13:30 local solar hour) and at night (at 1:30 local solar hour) for two Taiwan regions. Generally, we found the greatest daytime SUHIIs in summer and the smallest in the winter season, with 6.63 ± 1.33 °C and 3.43 ± 0.77 °C for northern Taiwan, and 4.21 ± 1.43 °C and 2.19 ± 0.92 °C for southern Taiwan, respectively. However, during the nighttime, the SUHII was more stable over seasons, as indicated by the considerably smaller variation than SUHIIs in the daytime for both Taiwan regions (Fig. 9). More specifically, at night, the SUHII was maximum in summer (1.57 ± 0.45 °C) and minimum in the winter season (0.47 ± 0.27 °C) for the northern part of Taiwan. In contrast, the nighttime SUHII averaged in the southern part of Taiwan reached a maximum of 1.12 ± 0.48 °C in spring and bottomed in winter (0.77 ± 0.38 °C) (Fig. 9).

In the day, the SUHIIs averaged for the northern region of Taiwan were higher than those for the southern part regardless of season ($p < 0.05$ for spring and summer, $p < 0.1$ for winter). However, the seasonal patterns of nighttime SUHIIs were more complex than daytime SUHIIs. In detail, the nighttime SUHIIs averaged in the southern region of Taiwan in spring and winter (1.12 ± 0.48 °C and 0.77 ± 0.38 °C, respectively) were higher as compared to their northern counterparts (0.99 ± 0.29 °C and 0.47 ± 0.27 °C, respectively). However, the cities located in Northern Taiwan exhibited more evident nighttime SUHIIs in summer and autumn (1.57 ± 0.45 °C and 1.17 ± 0.32 °C, respectively) as compared to the southern cities (1.06 ± 0.43 °C and 1.08 ± 0.36 °C, respectively) (Fig. 9).

The DNDs in Taiwan witnessed significant diurnal variations (Fig. 10). Overall, the largest and smallest DNDs were observed for both Taiwanese regions in the summer and winter, respectively. The DND averaged for Northern Taiwan was more significant compared to that for Southern Taiwan, with 5.06 ± 0.94 °C (in summer) and 2.96 ± 0.86 °C (in winter) for Northern Taiwan and 3.19 ± 1.08 °C (in summer) and 1.39 ± 0.71 °C (in winter) for Southern Taiwan, respectively (Fig. 10). Additionally, the annual DND averaged in Northern Taiwan (3.90 ± 0.91 °C) was higher compared with that in Southern Taiwan (2.35 ± 0.71 °C) (Fig. 10). As for SWD, the mean daytime SWD for all eleven Taiwanese cities was significantly greater than that during the night. More specifically, the mean SWDs were 2.56 ± 0.86 °C for the day and 0.66 ± 0.55 °C at night, respectively. Spatially, daytime SWDs were stronger than nighttime SWD in both regions, and higher SWDs were observed in Northern Taiwan compared to Southern Taiwan for the day and night (Fig. 10). Specifically, the daytime and nighttime SWDs were 3.21 ± 0.61 °C and 1.11 ± 0.47 °C in Northern Taiwan and 2.02 ± 0.64 °C and 0.29 ± 0.26 °C in Southern Taiwan, respectively.

Additionally, Fig. 11 shows the relationships across the studied cities between the daytime SUHII and nighttime SUHII (in summer, winter, and annually) and between summertime SUHII and wintertime SUHII (for day and night). This work reveals that daytime SUHII strongly correlated with SUHIIs at night in summer and annually ($r = 0.938$, $p < 0.001$ and $r = 0.757$, $p = 0.006$, respectively)

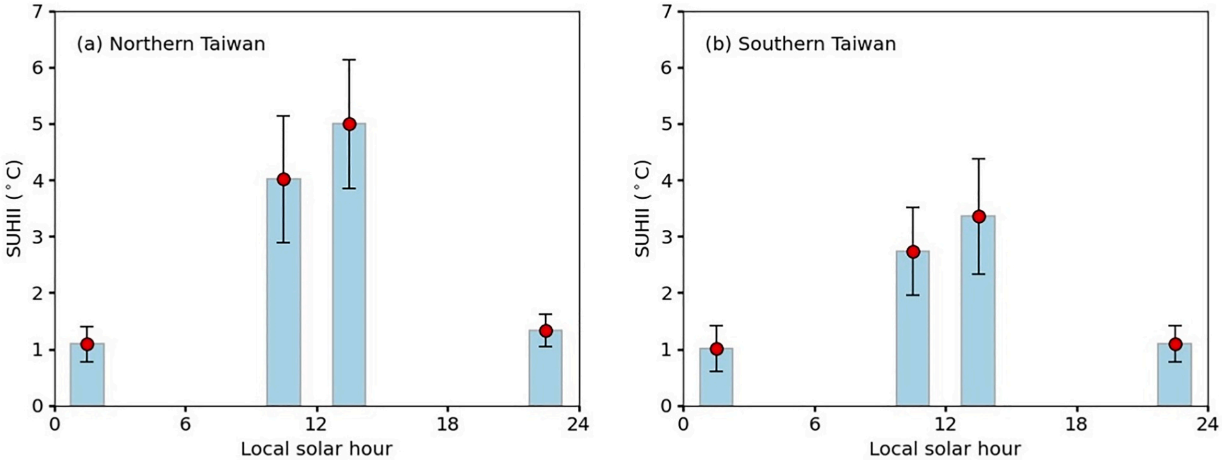

Fig. 7. The diurnal cycle of annual SUHII for the Taiwan regions, (a) Northern Taiwan and (b) Southern Taiwan (mean ± standard deviation).

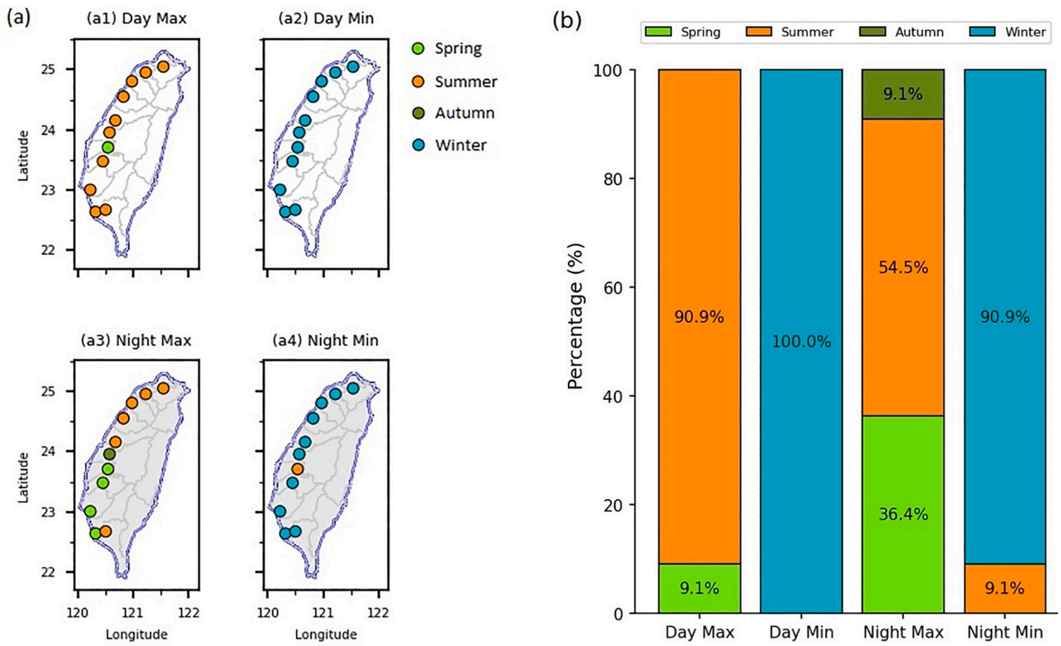

Fig. 8. The happening time of minimum or maximum SUHI intensity in the daytime and at night, (a) the spatial distribution, and (b) the percentage.

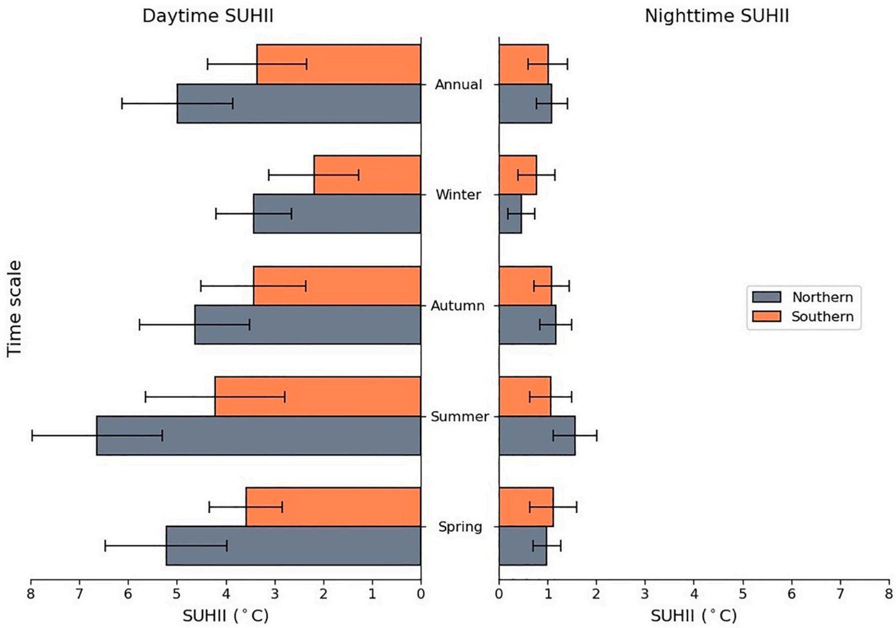

Fig. 9. The daytime and nighttime SUHIIs for different seasons in two Taiwan regions (mean ± standard deviation). Daytime SUHII and nighttime SUHII were calculated from the LST data at 13:30 and 1:30, respectively.

(Figs. 11(a), 11(c)), different from the findings of Peng et al. (2012) and Li et al. (2022), which observed no significant relationship between annual daytime and annual nighttime SUHII across cities. However, an insignificant relationship was found between winter nighttime SUHII and winter daytime SUHII ($r = -0.04$, $p = 0.891$) among cities (Fig. 11(b)). During the daytime, there was a strong relationship between summertime SUHII and wintertime SUHII ($r = 0.973$, $p < 0.001$), whereas the correlations were insignificant at night ($r = 0.130$, $p = 0.702$) (see Figs. 11(d), 11(e)).

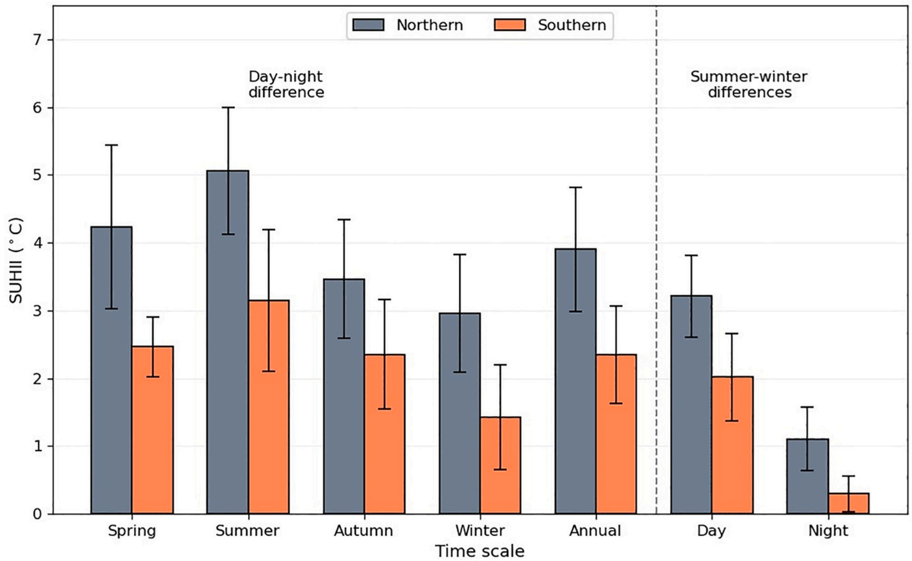

Fig. 10. DNDs and SWDs for two Taiwan regions (mean ± standard deviation). Daytime SUHII and nighttime SUHII were computed from the LST observations at 13:30 and 1:30, respectively.

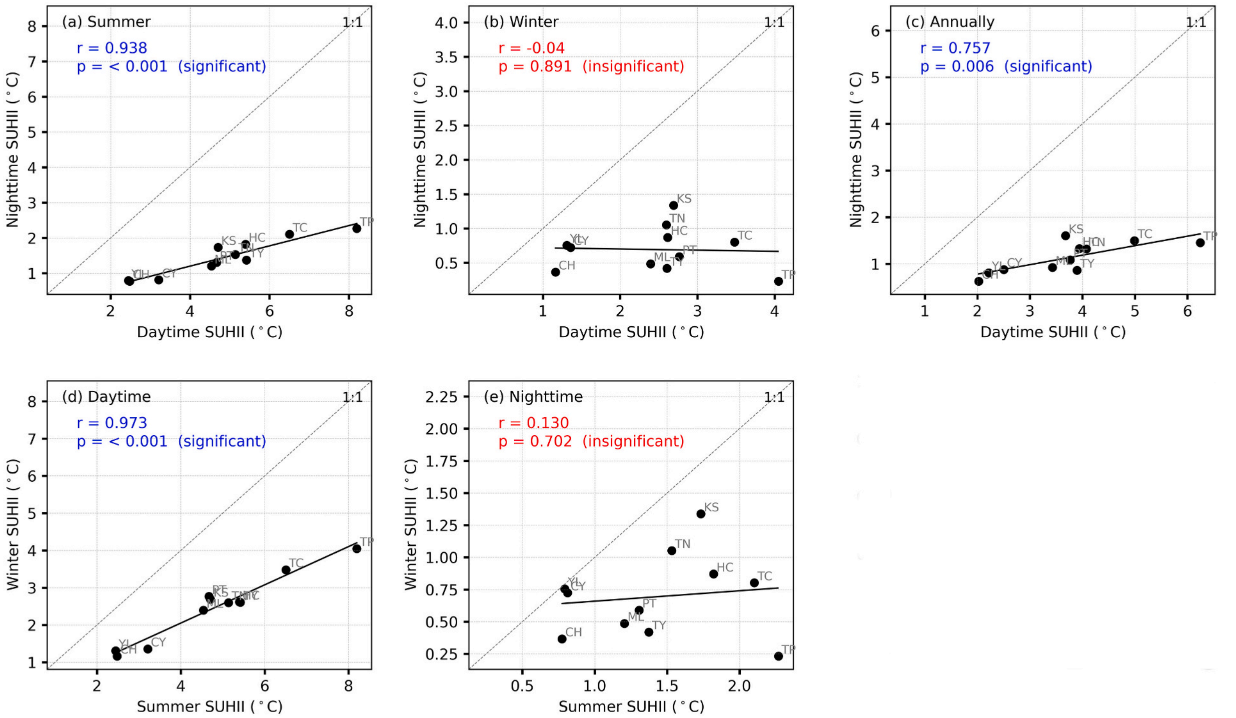

Fig. 11. Relationships between the daytime SUHII and nighttime SUHII in (a) summer, (b) winter, and (c) annually, and between summer SUHII and winter SUHII (d) in the daytime and (e) at night across eleven cities of Taiwan averaged over the period 2003–2020.

4.2. Relationships between SUHII and its drivers

4.2.1. Effects of each associated determinant of SUHII

During the day, the SUHIIs had negative and significant correlations with the urban-rural contrasts in NDLI (ΔNDLI) and Enhanced Vegetation Index (ΔEVI) for all seasons (Table 3). Also, we observed statistically significant and positive correlations of urban-rural differences in Built-up Intensity (ΔBI) and Nighttime Lights (ΔNL) with daytime SUHII regardless of the season. Additionally, we found that the daytime SUHIs positively correlated with the urban-rural contrasts in Aerosol Optical Depth (ΔAOD), White Sky Albedo (ΔWSA), and total accumulated precipitation (MAP) (Table 3). However, at night, the SUHII showed positive correlations with mean air temperature (MAT) for all seasons. Moreover, we found significant positive correlations of ΔBI and ΔNL with nighttime SUHI in the summer and autumn. In addition, SUHII in the nighttime demonstrated negative correlations with ΔEVI and ΔNDLI, but was only statistically significant in summer, autumn, and annually. We also observed that the nighttime SUHII significantly and negatively correlated with the ΔWSA and MAP in winter (see Table 3).

4.2.2. *Combined effects of factors controlling SUHII*

The stepwise multiple regression model results indicated that a greater fraction of the spatial SUHII variabilities during the day than at nighttime can be explained by the variables considered in this work, regardless of the season. Also, while the largest daytime total explanation rate occurred in summer (87.0%), followed by winter (85.3%), spring (81.5%), and autumn (79.5%), the order for nighttime total explanation rate was summer (77.0%), autumn (68.7%), spring (50.3%), and winter (44.9%) (Table 3). In the day, the urban-rural contrast in NDLI (ΔNDLI) explained a significantly large fraction of SUHII variation in summer (81.2%), autumn (74.9%), and winter (77.7%). In contrast, ΔBI made up most of the spatial variation of daytime SUHII in the spring season (67.6%). However, the nighttime SUHII's spatial variability was mainly explained by nighttime light signals (ΔNL) in summer (65.3%) and autumn (54.5%), by air temperature in spring (25.9%), and by precipitation in winter (21.8%) (Table 3). In addition, for the annual scale, while ΔNDLI accounted for almost the variation of daytime SUHII (83.1%), the variation of nighttime SUHII was primarily explained by ΔNL (58.5%) (Table 3).

5. Discussion

5.1. *Seasonal and diurnal cycles of SUHII in Taiwan*

5.1.1. *Diurnal cycles of SUHII*

This study suggests that the SUHI intensities significantly fluctuated in a diurnal variation, with similar patterns but dissimilar magnitudes in the two regions of Taiwan (see Fig. 7). The diurnal variation is due to the day-night difference in heat sources (Oke, 1982; Peng et al., 2012; Zhou et al., 2014b). In the daytime, the SUHI is widely contributed by decreased latent heat flux, strengthening the sensible heat flux and upward thermal radiation in urban areas (Shepherd, 2005; Zhou et al., 2019). In contrast, at night, the SUHI is mainly caused by greater ground heat storage, which is trapped during the daytime and later emitted at night (Arnfield, 2003; Cao et al., 2016; Clinton and Gong, 2013; Voogt and Oke, 2003). In addition, anthropogenic heat production can increase the SUHI by increasing heat storage and upwelling thermal radiance (Rizwan et al., 2008).

Our results reveal that the daytime SUHIIs were higher than the nighttime SUHIIs in both regions of Taiwan. Our findings were similar to the observations made in America by Imhoff et al. (2010). However, these patterns slightly differed from the outcomes of those of a previous study conducted in China, which illustrated that the average annual daytime SUHII was less as compared to that at night in the northern region, but the opposite observations were found in its southeast region (Zhou et al., 2016b; Zhou et al., 2014b).

In addition, this work observed significantly higher SUHIIs for northern Taiwan than those for its southern counterparts, regardless

Table 3
Correlation coefficients of the SUHII with eight potential drivers *(ΔEVI, ΔNDLI, ΔWSA, ΔAOD, ΔBI, ΔNL, MAT, and MAP),* along with independent explanation power (shown in parentheses as a percentage) of each factor obtained through stepwise regression models.

		Factors								Overall R^2
Diurnal	Seasonal	ΔNDLI	ΔEVI	ΔWSA	ΔAOD	ΔBI	ΔNL	MAT	MAP	
Day	Spring	−0.83*** (2.0)	−0.76***	0.73** (9.8)	0.41	0.89*** (67.6)	0.76***	−0.25 (2.1)	0.63**	– (81.5)
	Summer	−0.93*** (81.2)	−0.91***	0.47	0.42	0.86*** (5.1)	0.79*** (0.7)	0.54*	0.01	– (87.0)
	Autumn	−0.90*** (74.9)	−0.83***	0.52*	0.50 (0.4)	0.80***	0.80*** (1.8)	0.18	0.71** (2.4)	– (79.5)
	Winter	−0.93*** (77.7)	−0.82***	0.63** (1.3)	0.67** (4.3)	0.78*** (0.7)	0.73** (0.4)	0.05 (0.6)	0.51 (0.3)	– (85.3)
	Annual	−0.93*** (83.1)	−0.85***	0.65** (0.6)	0.49 (2.5)	0.85*** (1.2)	0.79***	0.04 (0.7)	0.61** (0.9)	– (89.0)
Night	Spring	0.05 (2.6)	−0.32 (2.5)	−0.03	0.34	0.41 (18.1)	0.56*	0.63** (25.9)	−0.13 (1.2)	– (50.3)
	Summer	−0.86*** (10.5)	−0.86***	0.31	0.54* (1.2)	0.83***	0.88*** (65.3)	0.52*	0.16	– (77.0)
	Autumn	−0.79*** (2.0)	−0.75***	0.23	0.74*** (10.2)	0.69**	0.85*** (54.5)	0.56* (2.0)	0.56*	– (68.7)
	Winter	0.00	−0.19 (2.8)	−0.58* (8.5)	0.20	0.13 (11.8)	0.24	0.58*	−0.59* (21.8)	– (44.9)
	Annual	−0.65** (2.5)	−0.70**	0.11 (1.3)	0.73** (12.4)	0.71**	0.82*** (58.5)	0.56*	0.67**	– (74.7)

*** $p < 0.01$, p: significance level (*p*-value); ** $0.01 \le p < 0.05$; * $0.05 \le p < 0.1$.

of seasonality in the daytime. The reason for the difference might be linked to different patterns of LULC. Specifically, the surrounding rural regions of Northern Taiwan were surrounded by a higher fraction of forests as compared to those of Southern Taiwan (covered mainly by agricultural land (mostly cropland)) (Fig. 1 and Fig. 12). We found higher negative ΔEVI of Northern Taiwan than the Southern counterparts, regardless of seasonality (Fig. 13). Additionally, the greater surface albedo of cropland (compared to the forest) can decrease the heat stored in the day (in rural regions) and then support lower nighttime rural temperatures, thus strengthening the SUHII at night (Jin et al., 2005; Zhou et al., 2016b; Zhou et al., 2014b).

5.1.2. Seasonal cycles of SUHII

Our findings demonstrate the significant seasonal variation of SUHIIs across eleven Taiwanese cities during the daytime and nighttime (Figs. 6 and 9). Specifically, in the daytime, the SUHIIs were smallest in winter and strongest in summer for both regions of Taiwan, mainly owing to the lowest and greatest cooling effects of vegetation in rural regions during the two seasons (Clinton and Gong, 2013; Imhoff et al., 2010; Peng et al., 2012; Zhou et al., 2016b). The daytime SWD in northern Taiwan was higher than in southern Taiwan (Fig. 10), mainly owing to the higher variation of seasonal vegetation activity in northern Taiwan compared to southern Taiwan. More specifically, the negative summer-winter difference of ΔEVI in northern Taiwan was about twice that in southern Taiwan (Fig. 13). We also observed that the nighttime SUHIIs were the weakest during winter for both parts of Taiwan (Fig. 10). This may be partly due to the lowest air temperature and relatively high positive ΔWSA in winter (Fig. 13). Interestingly, most cities in the southern part of Taiwan (4/6) experienced the largest nighttime SUHIIs in spring (not in summer as cities in the northern part) (Fig. 8), which may be partly owing to the highest precipitation in summer and agricultural practices in southern Taiwan, resulting in decreasing summertime SUHII at night (Table 3).

Our results suggest that the SWD for all eleven Taiwanese cities was more evident during the day than at night. The significant contrast stemmed from the fact that the summertime SUHII was more evident than the wintertime one for almost all cities (except Yunlin) during the day. The same pattern was observed for most cities at night, but with a smaller difference between summer and winter. Furthermore, the seasonal changes of nighttime SUHIIs were weaker than those in the day for both regions of Taiwan (Fig. 10). The reason for this is because the surface heat storage, which does not vary substantially among seasons in comparison with the vegetation activity (the main factor affecting daytime SUHII), is a primary source of energy responsible for nighttime SUHII (Oke, 1982; Zhou et al., 2016b).

5.2. The possible driving mechanisms of SUHII's spatial variations

5.2.1. Vegetation and NDLI

The presence of vegetation is known to strengthen latent heat fluxes, which can produce a cooling effect on LST and help alleviate the SUHI impacts (Li et al., 2022; Li et al., 2019; Peng et al., 2012; Tran et al., 2006; Zhou et al., 2016b). Our present study supports this mechanism since daytime SUHIIs negatively and significantly correlated to ΔEVI across all seasons (Table 3). The effect of vegetation during winter daytime ($r = -0.82$, $p < 0.01$) was lower than during summer daytime ($r = -0.91$, $p < 0.01$) primarily due to the lower vegetation activities (less latent heat flux) during winter. Although the absence of vegetation transpiration at night would suggest weak

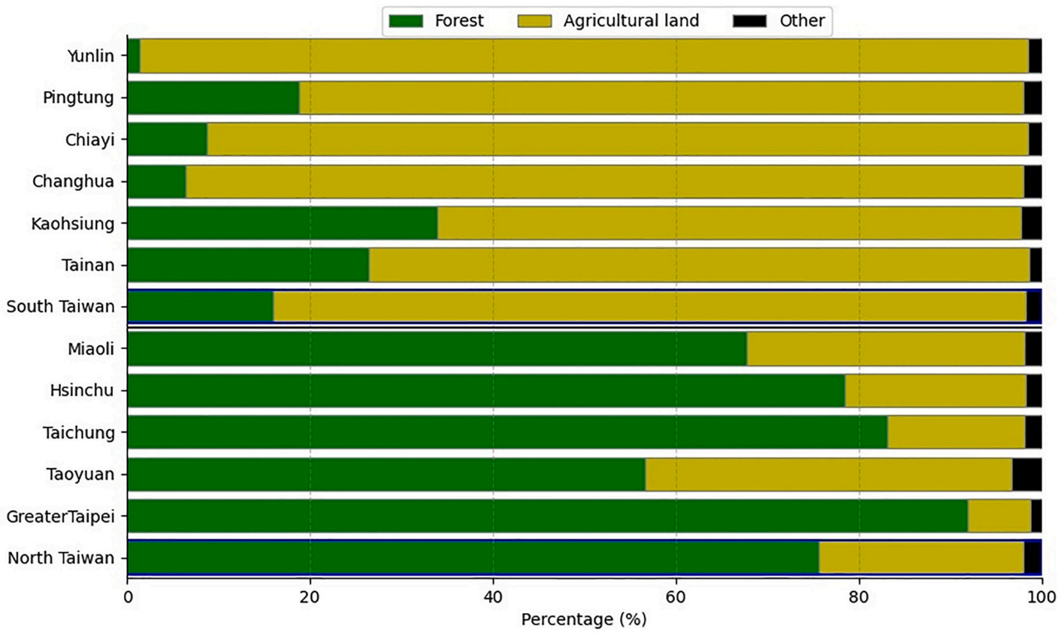

Fig. 12. Percentage of LULC in rural area for each city.

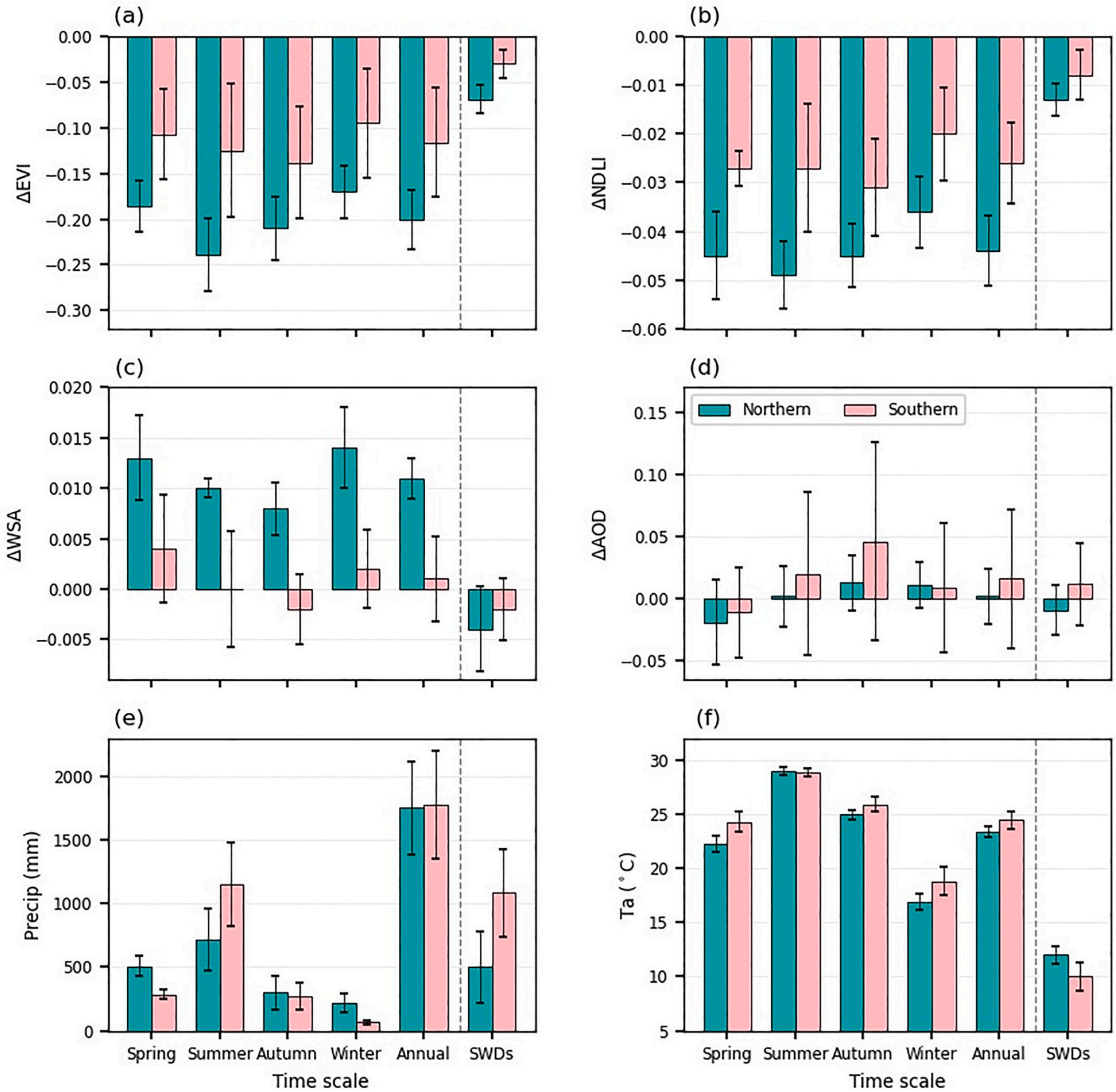

Fig. 13. Factor values by season for two Taiwan regions.

correlations between SUHIIs and ΔEVI across seasons, significant negative relationships between ΔEVI and SUHII were found during summer ($r = -0.86$, $p < 0.01$) and autumn ($r = -0.75$, $p < 0.01$) nights. The reason for this may be caused by the fact that the ΔEVI were highly collinearly correlated with factors such as ΔNL and ΔISA, which controlled nighttime SUHII in summer and autumn.

In this work, the NDLI was utilized as an indicator of the evapotranspiration of vegetation (Liou et al., 2018). The process of evaporation and transpiration from water surfaces and vegetation can generate a cooling effect on LST, thereby mitigating the impact of SUHI (Mohammad and Goswami, 2021; Peng et al., 2012; Shastri et al., 2017). The outcomes presented in Table 3 confirm this mechanism since SUHII was found to be significantly and negatively correlated to ΔNDLI in the day across all seasons, with stronger correlations during the daytime compared to nighttime.

5.2.2. *White sky albedo and aerosol optical depth*

Overall, urbanization can reduce surface albedo for pre-urban areas with high surface albedo, primarily due to the urban canopy and construction components utilized for roads and buildings in urban regions (Karimi et al., 2020). Low-albedo materials in urban areas can absorb more energy and increase net all-wave radiation and heat storage, thus reducing heat loss (Arnfield, 2003; Oke, 1982; Stewart and Oke, 2012; Zhou et al., 2014b). The city with higher negative ΔWSA is expected to have higher nighttime SUHII because the ground heat fluxes, which stem from energy storage in the daytime, are the primary source of heat leading to SUHII in the nighttime

(Arnfield, 2003; Shepherd, 2005). Our research verifies this mechanism during winter because we found that the nighttime SUHII was strongly and negatively correlated with ΔWSA in this season (Table 3).

Generally, aerosols have a cooling effect on the ground surface by reducing the volume of shortwave radiance contacting the ground surface, thus decreasing LST. However, under certain circumstances, aerosols can strengthen LST due to their higher efficiency in absorbing and releasing radiance in the longwave atmospheric spectrum compared to water vapor and greenhouse gases, thus potentially increasing the longwave radiation energy (Cao et al., 2016; Jacobson, 1998). Therefore, the combined effect relies on the particles' initial size and growth because of aging and absorption of water vapor (Rudich et al., 2007). In this work, ΔAOD witnessed a positive correlation with mean annual SUHII at night in eleven cities in Taiwan ($r = 0.73$, $p < 0.01$), and ΔAOD was not significantly correlated with annual and summer daytime SUHI intensities (Table 3), which is similar to observations from previous research (Cao et al., 2016). In addition, we observed that ΔAOD is significantly positively correlated to daytime SUHII ($r = 0.67$, $p < 0.05$) in winter, as well as nighttime SUHII in autumn ($r = 0.74$, $p < 0.01$).

5.2.3. *Built-up intensity and nighttime lights*

The urban region with greater build-up (BI) is known to retain higher surface temperatures than the surrounding rural region with lower BI. The reason is that urban building materials, street canyons, and anthropogenic activities can trap more heat, and a reduction in vertical heat flux change leads to less heat loss (Mohammad and Goswami, 2021; Oke, 1982; Yow, 2007; Zhou et al., 2014a). A city with a larger ΔBI is expected to experience a more significant SUHI intensity, especially in the nighttime when the ground heat store is emitted (Li et al., 2011; Tran et al., 2006; Zhou et al., 2014b). Our study partly confirms this mechanism because ΔBI had a significant positive correlation with nighttime SUHI intensities during summer ($r = 0.83$, $p < 0.01$) and autumn ($r = 0.69$, $p < 0.05$) while showing insignificant correlations with nighttime SUHI intensities in spring ($r = 0.41$, $p > 0.1$) and winter ($r = 0.13$, $p > 0.1$) (Table 3). Furthermore, ΔBI significantly correlated with SUHIIs during the daytime across all seasons.

The anthropogenic heat release indirectly affects SUHI by converting it into sensible heat flux or other energy components (Peng et al., 2012; Zhou et al., 2019; Zhou et al., 2014b). Previous studies used nighttime light signals (NL) to surrogate anthropogenic heat releases (Li et al., 2022; Li et al., 2019; Mohammad and Goswami, 2021; Peng et al., 2012). Our research found that ΔNL had significantly positive correlations with SUHII in the daytime (for all seasons) and in the nighttime (for summer, autumn, and spring) (Table 3). The positive correlation between SUHII and ΔNL in our study is in line with the results from other studies (Li et al., 2019; Mohammad and Goswami, 2021; Zhou et al., 2014b).

5.2.4. *Climatic factors*

Atmospheric temperatures can not only influence LST in urban and surrounding rural areas (Wu et al., 2013), but also regulate other factors, such as vegetative activity (Piao et al., 2003), surface albedo (Hall, 2004), and anthropogenic heat release (Santamouris et al., 2001). These factors can directly or indirectly control SUHII. In this study, we found that nighttime SUHI intensity was positively and significantly correlated with air temperature for all seasons (Table 3), which differs from previous studies that reported significant negative relationships between SUHI intensity and air temperature at night in winter, summer, and annually (Peng et al., 2012; Zhou et al., 2016b; Zhou et al., 2014b). Moreover, our results showed insignificant correlations between air temperature and daytime SUHII in most seasons (apart from in summer, $r = 0.54$, $p < 0.1$), different from significant positive relationships between air temperature and daytime SUHII for 32 major cities in China (Zhou et al., 2016b; Zhou et al., 2014b).

The influence of climate conditions on SUHI can also be more directly explained through soil moisture information (Oke, 1982; Tran et al., 2006; Zhou et al., 2016b). Higher soil moisture in rural compared to urban areas can decrease SUHII at night, but enhance daytime SUHII because of the rise in thermal inertia and heat capacity of soil moisture, respectively (Yao et al., 2018b; Zhou et al., 2016b; Zhou et al., 2014b). In our research, total precipitation (MAP) showed significant positive correlations with daytime SUHII in spring ($r = 0.63$, $p < 0.05$), autumn ($r = 0.71$, $p < 0.05$), and annually ($r = 0.61$, $p < 0.05$). However, at night, total precipitation was significantly positively correlated to SUHII during autumn ($r = 0.56$, $p < 0.1$) and annually ($r = 0.67$, $p < 0.05$) while having a significant negative correlation with wintertime SUHII ($r = -0.59$, $p < 0.1$) (Table 3).

5.2.5. *Combined effects*

Generally, the combined impacts of the factors examined in this study provide a better explanation for daytime SUHIIs than for nighttime ones across all seasons, with explanatory rates varying from 44.9% for nighttime in spring to 87.0% for daytime in summer (Table 3). Our findings indicate that the underlying mechanisms of spatial patterns of nighttime SUHIIs were more complicated, especially in winter (44.9%) and spring (50.3%), which is similar to the results of Li et al. (2019) but different from the findings of Zhou et al. (2014b). Therefore, other factors, such as LULC and landscape characteristics, may significantly impact the distribution of SUHII (Li et al., 2011; Oke, 1982).

Previous studies indicated that vegetation is crucial in controlling summer daytime SUHII variability (Peng et al., 2012; Yuan and Bauer, 2007; Zhou et al., 2016b). In addition, Zhou et al. (2014b) reported that anthropogenic heat releases, strongly correlated with vegetation activity, explained most spatial variations in daytime SUHII during summer. Our findings suggest that ΔNDLI, which was closely related to vegetation activity, can account for a significant portion of the spatial variability in daytime SUHII during summer (81.2%), autumn (74.9%), winter (77.7%), and annually (84.4%). In spring, built-up density explained about 67.6% of daytime SUHII variation.

However, anthropogenic heat emissions could explain most of the nighttime SUHII variability in summer (65.3%) and autumn (54.5%). In contrast, the main contributor to the nighttime SUHII spatial variability was the air temperature in spring (25.9%) and precipitation in winter (21.8%). These findings differ from previous studies (Peng et al., 2012; Zhou et al., 2014b), suggesting that albedo explained most nighttime SUHII variability in summer and wintertime.

5.3. Limitations and future works

The present study has several limitations, which should be pointed out. First, the Taiwan area is approximately 36,193 km^2; therefore, the MODIS data with a 1 km spatial resolution may not be optimal for characterizing spatio-temporal patterns of SUHI for Taiwanese cities. Second, the period of data from 2003 to 2020 is not long, and we did not assess changes in SUHI through time. Third, due to the cloud influences, we have yet to analyze all Taiwanese cities, particularly the eastern ones. Fourth, we attempted to reduce the effects of water and elevation on SUHI analysis, but coastal effects may remain. For future work, several reconstructions of missing values and image fusion methods can be applied for MODIS and Landsat data to improve satellite image continuity and spatial resolution, thus allowing better SUHII studies through time. Besides, other factors, which can affect the SUHI, such as wind speed, wind direction, and solar radiation, should be included in the SUHI analysis.

6. Conclusions

In conclusion, this research is the first to comprehensively investigate the diurnal, seasonal, and spatial pattern of SUHII and their associated driving factors in eleven cities in Taiwan using remotely sensed data and climate information from 2003 to 2020. Also, for the first time, we integrated the NDLI, which served as an indicator to characterize the potential latent heat flux of the Earth's surface, into the SUHI study. This work provides a better understanding of the spatio-temporal variations and possible underlying mechanisms of the SUHI phenomenon at a regional scale.

The results revealed that daytime SUHIIs are consistently more intense than nighttime SUHIIs in all seasons and most cities, with higher intensities witnessed in northern cities compared to southern cities, apart from springtime and wintertime at night. Regarding seasons, the SUHII changes significantly, with more variation during the day than at night. Also, our results showed that while the SUHII is maximized in the summertime and minimized in wintertime for most cities, the strongest nighttime SUHII occurs in spring rather than summer, as previously assumed for most cities in Southern Taiwan. By using Pearson's correlation analysis, during the daytime, we found that the factors that have negative and significant correlations with SUHII across all seasons include vegetative activity (ΔEVI) and ΔNDLI. In contrast, built-up intensity (ΔBI) and anthropogenic heat emissions (ΔNL) are positively correlated with daytime SUHIIs. At night, while SUHIIs exhibit positive correlations with air temperature, ΔBI, and ΔNL, they correlate negatively with ΔEVI, ΔNDLI, and ΔWSA for some seasons. By employing the stepwise regression technique, we found that the factors examined in this work can explain a considerably higher portion of the variation of SUHII during the daytime than the nighttime, which suggests the underlying mechanisms of the spatial patterns of SUHII are more complicated at night, especially in the spring and winter seasons.

This paper emphasizes the importance of studying spatiotemporal variations of SUHII and the associated driving forces, which helps to understand the SUHII's mechanism over space and time without complex modeling. This information may help researchers and government agencies assess and adapt the SUHI effects. The mitigation of SUHI can be achieved by increasing urban vegetation activity and water surfaces in urban sites and reducing anthropogenic heat emissions.

CRediT authorship contribution statement

Yuei-An Liou: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Duy-Phien Tran:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Kim-Anh Nguyen:** Formal analysis, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Appendix

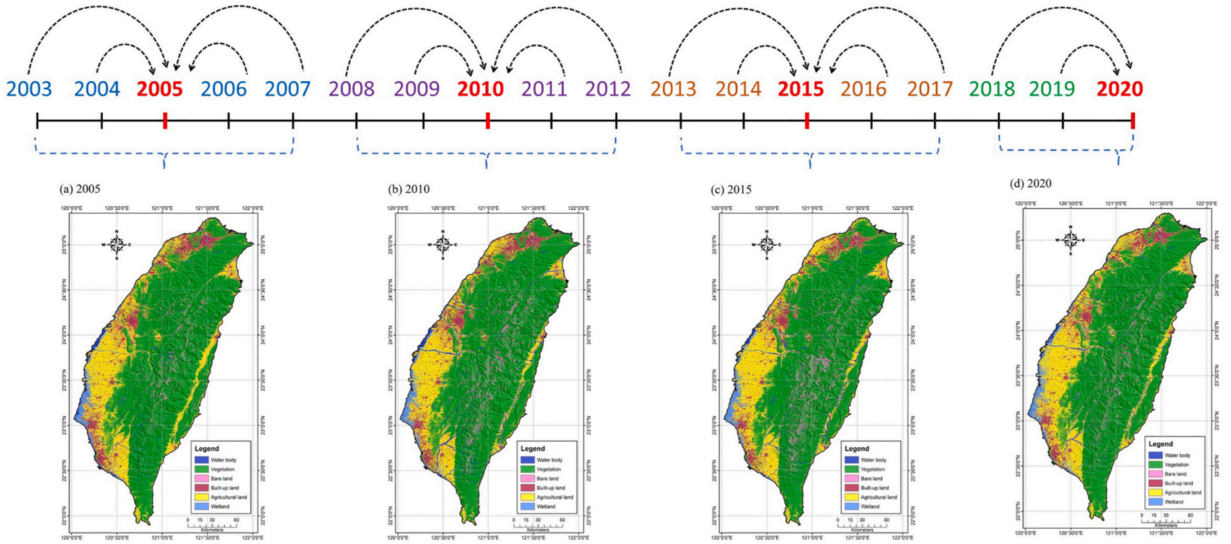

Fig. A1. Taiwan LULC maps for four years (2005, 2010, 2015, and 2020) and the corresponding periods: 2003–2007, 2008–2012, 2013–2017, and 2018–2020.

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